

Chapter 20

The AA-DD Model

Ideally, it would be nice to develop a way to keep track of all the cause-and-effect relationships that are presumed to exist at the same time. From the previous chapters it is clear, for example, that the money supply affects the interest rates in the money market, which in turn affects the exchange rates in the foreign exchange (Forex) market, which in turn affects demand on the current account in the goods and services (G&S) market, which in turn affects the level of GNP, which in turn affects the money market, and so on. The same type of string of repercussions can be expected after many other changes that might occur. Keeping track of these effects and establishing the final equilibrium values would be a difficult task if not for a construction like the AA-DD model. This model merges the money market, the Forex market, and the G&S market into one supermodel. The construction of the AA-DD model is presented in this chapter.

20.1 Overview of the AA-DD Model

LEARNING OBJECTIVE

1. Understand the basic structure and results of the AA-DD model of national output and exchange rate determination.

This chapter describes the derivation and the mechanics of the **AA-DD model**. The AA-DD model represents a synthesis of the three previous market models: the foreign exchange (Forex) market, the money market, and the goods and services market. In a sense, there is really very little new information presented here. Instead, the chapter provides a graphical approach to integrate the results from the three models and to show their interconnectedness. However, because so much is going on simultaneously, working with the AA-DD model can be quite challenging.

The AA-DD model is described with a diagram consisting of two curves (or lines): an **AA curve** representing asset market equilibriums derived from the money market and foreign exchange markets and a **DD curve** representing goods market (or demand) equilibriums. The intersection of the two curves identifies a market equilibrium in which each of the three markets is simultaneously in equilibrium. Thus we refer to this equilibrium as a **superequilibrium**.

Results

The main results of this section are descriptive and purely mechanical. The chapter describes the derivation of the AA and DD curves, explains how changes in exogenous variables will cause shifts in the curves, and explains adjustment from one equilibrium to another.

- a. The DD curve is the set of exchange rate and GNP combinations that maintain equilibrium in the goods and services market, given fixed values for all other exogenous variables.
- b. The DD curve shifts rightward whenever government demand (G), investment demand (I), transfer payments (TR), or foreign prices ($P_{\mathcal{E}}$) increase or when taxes (T) or domestic prices ($P_{\$}$) decrease. Changes in the opposite direction cause a leftward shift.
- c. The AA curve is the set of exchange rate and GNP combinations that maintain equilibrium in the asset markets, given fixed values for all other exogenous variables.
- d. The AA curve shifts upward whenever money supply (M^S), foreign interest rates ($i_{\mathcal{E}}$), or the expected exchange rate ($E_{\$/\mathcal{E}}$) increase or when domestic prices ($P_{\$}$) decrease. Changes in the opposite direction cause a downward shift.
- e. The intersection of the AA and DD curves depicts a superequilibrium in an economy since at that point the goods and services market, the domestic money market, and the foreign exchange market are all in equilibrium simultaneously.



cause a shift of the AA or DD curves and move the economy out of equilibrium, temporarily.

Adjustment to a new equilibrium follows the principle that adjustment in the asset markets occurs much more rapidly than adjustment in the goods and services market. Thus adjustment to the AA curve will always occur before adjustment to the DD curve.

Connections

The AA-DD model will allow us to understand how changes in macroeconomic policy—both monetary and fiscal—can affect key aggregate economic variables when a country is open to international trade and financial flows while accounting for the interaction of the variables among themselves. Specifically, the model is used to identify potential effects of fiscal and monetary policy on exchange rates, trade balances, GDP levels, interest rates, and price levels both domestically and abroad. In subsequent chapters, analyses will be done under both floating and fixed exchange rate regimes.

KEY TAKEAWAYS

- The AA-DD model integrates the workings of the money-Forex market and the G&S model into one supermodel.
- The AA curve is derived from the money-Forex model. The DD curve is derived from the G&S model.
- The intersection of the AA and DD curves determines the equilibrium values for real GNP and the exchange rate.
- Comparative statics exercises using the AA-DD model allow one to identify the effects of changes in exogenous variables on the level of GDP and the exchange rate, while assuring that the Forex, the money market, and the G&S market all achieve simultaneous equilibrium.

EXERCISE

1. **Jeopardy Questions.** As in the popular television game show, you are given an answer to a question and you must respond with the question. For example, if the answer is “a tax on imports,” then the correct question is “What is a tariff?”
 - a. At the intersection of the AA and DD curves, the goods and services market, the money market, and this market are simultaneously in equilibrium.
 - b. The term used to describe the type of equilibrium at the intersection of the AA curve and the DD curve.

20.2 Derivation of the DD Curve

LEARNING OBJECTIVE

1. Learn how to derive the DD curve from the G&S model.

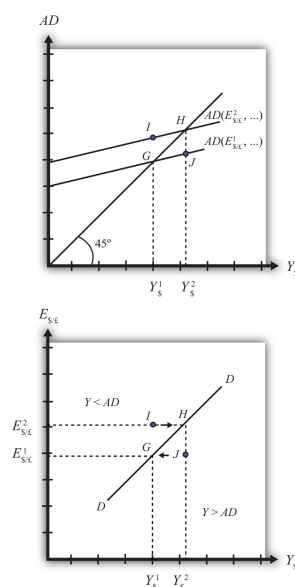
The DD curve is derived by transferring information described in the goods and services (G&S) market model onto a new diagram to show the relationship between the exchange rate and equilibrium gross national product (GNP). The original G&S market, depicted in the top part of [Figure 20.1 "Derivation of the DD Curve"](#), plots the aggregate demand (AD) function with respect to changes in U.S. GNP ($Y_{\$}$). Aggregate demand is measured along the vertical axis and aggregate supply (or the GNP) is measured on the horizontal axis. As discussed in [Chapter 19 "National Output Determination"](#), [Section 19.7 "The Aggregate Demand Function"](#), the AD function is dependent upon several different exogenous variables, most notably the exchange rate between domestic and foreign currency ($E_{\$/\pounds}$). However, AD is also affected by investment demand (I), government demand (G), government tax revenues (T), government transfer payments (TR), and the price level in the domestic

20.1 G&S Market (for a quick reference.) In this exercise, since our focus is on the exchange rate, we label the AD function in Figure 20.1 "Derivation of the DD Curve" as $AD(E_{\$/\pounds}, \dots)$, where the ellipsis (...) is meant to indicate there are other unspecified variables that also influence AD.

Table 20.1 G&S Market

Exogenous Variables	$E_{\$/\pounds}, I, G, T, TR, P_{\$}, P_{\pounds}$
Endogenous Variable	$Y_{\$}$

Figure 20.1 Derivation of the DD Curve



Initially, let's assume the exchange rate is at a value in the market given by $E_{\$/\pounds}^1$. We need to remember that all the other variables that affect AD are also at some initial level. Written explicitly, we could write AD as $AD(E_{\$/\pounds}^1, I^1, G^1, T^1, TR^1, P_{\$}^1, P_{\pounds}^1)$. The AD function with exchange rate $E_{\$/\pounds}^1$ intersects the forty-five-degree line at point G which determines the equilibrium level of GNP given by $Y_{\1 . These two values are transferred to the lower diagram at point G determining one point on the DD curve ($Y_{\$}^1, E_{\$/\pounds}^1$).

Next, suppose $E_{\$/\pounds}$ rises from $E_{\$/\pounds}^1$ to $E_{\$/\pounds}^2$, ceteris paribus. This corresponds to a depreciation of the U.S. dollar with respect to the British pound. The ceteris paribus assumption means that investment, government, taxes, and so on stay fixed at levels I^1, G^1, T^1 , and so on. Since a dollar depreciation makes foreign G&S relatively more expensive and domestic goods relatively cheaper, AD shifts up to $AD(E_{\$/\pounds}^2, \dots)$. The equilibrium shifts to point H at a GNP level $Y_{\2 . These two values are transferred to the lower diagram at point H, determining a second point on the DD curve ($Y_{\$}^2, E_{\$/\pounds}^2$).

The line drawn through points G and H on the lower diagram is called the DD curve. The DD curve plots an equilibrium GNP level for every possible exchange rate that may prevail, ceteris paribus. Stated differently, the DD curve is the combination of exchange rates and GNP levels that maintain equilibrium in the G&S market, ceteris paribus. We can think of it as the set of aggregate demand equilibriums.

A Note about Equilibriums

An equilibrium in an economic model typically corresponds to a point toward which the endogenous variable values will converge based on some behavioral assumption about the participants in the model. In this case, equilibrium is not represented by a single point. Instead every point along the DD curve is an equilibrium value.

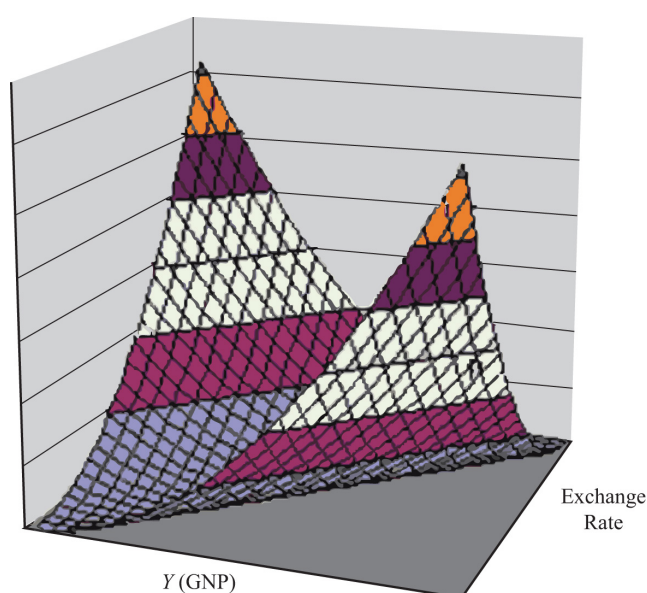
If the economy were at a point above the DD curve, say, at I in the lower diagram, the exchange rate

At point I , read on the vertical axis. In the G&S model, whenever aggregate demand exceeds aggregate supply, producers respond by increasing supply, causing GNP to rise. This continues until $AD = Y$ at point H . For all points to the left of the DD curve, $AD > Y$, therefore the behavior of producers would cause a shift to the right from any point like I to a point like H on the DD curve.

Similarly, at a point such as J , to the right of the DD curve, the exchange rate is $E_{\$/\text{£}}^1$ and the GNP level is at $Y_{\text{\$}}^2$. This corresponds to point J in the upper diagram above where aggregate demand is less than supply ($AD < Y$). In the G&S model, whenever supply exceeds demand, producers respond by reducing supply, thus GNP falls. This continues until $AD = Y$ at point G . For all points to the right of the DD curve, $AD < Y$, therefore the behavior of producers would cause a shift to the left from any point like J to a point like G on the DD curve.

A useful analogy is to think of the DD curve as a river flowing through a valley. (See the 3-D diagram in Figure 20.2 "A 3-D DD Curve".) The hills rise up to the right and left along the upward-sloping DD curve. Just as gravity will move a drop of water downhill onto the river valley, firm behavior will move GNP much in the same way: right or left to the lowest point along the DD curve.

Figure 20.2 A 3-D DD Curve



KEY TAKEAWAYS

- The DD curve plots an equilibrium GNP level for every possible exchange rate that may prevail, *ceteris paribus*.
- Every point on a DD curve represents an equilibrium value in the G&S market.
- The DD curve is positively sloped because an increase in the exchange rate (meaning a decrease in the U.S. dollar value) raises equilibrium GNP in the G&S model.

EXERCISE

1. **Jeopardy Questions.** As in the popular television game show, you are given an answer to a question and you must respond with the question. For example, if the answer is "a tax on imports," then the correct question is "What is a tariff?"
 - a. This is what has happened to its currency value if an economy's exchange rate and GNP combination moves upward along an upward-sloping DD curve.
 - b. Of *greater than*, *less than*, or *equal to*, this is how aggregate demand compares to GNP when the economy has an exchange rate and GNP combination that places it to the left of the DD curve.
 - c. Of *greater than*, *less than*, or *equal to*, this is how aggregate demand compares to GNP when the economy has an exchange rate and GNP combination that

20.3 Shifting the DD Curve

LEARNING OBJECTIVE

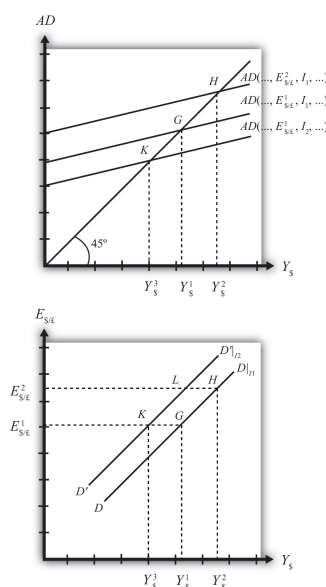
1. Learn which exogenous variables will shift the DD curve and in which direction.

The DD curve depicts the relationship between changes in one exogenous variable and one endogenous variable within the goods and services (G&S) market model. The exogenous variable assumed to change is the exchange rate. The endogenous variable affected is the gross national product (GNP). At all points along the DD curve, it is assumed that all other exogenous variables remain fixed at their original values.

The DD curve will shift, however, if there is a change in any of the other exogenous variables. We illustrate how this works in [Figure 20.3 "DD Curve Effects from a Decrease in Investment Demand"](#). Here, we assume that the level of investment demand in the economy falls from its initial level I_1 to a lower level I_2 .

At the initial investment level (I_1) and initial exchange rate ($E_{\$/\text{£}}^1$) the AD curve is given by $AD(..., E_{\$/\text{£}}^1, I_1, ...)$. The AD curve intersects the forty-five-degree line at point G , which is transferred to point G on the DD curve below. If the investment level and all other exogenous variables remain fixed while the exchange rate increases to $E_{\$/\text{£}}^2$, then the AD curve shifts up to $AD(..., E_{\$/\text{£}}^2, I_1, ...)$, generating the equilibrium points H in both diagrams. This exercise plots out the initial DD curve labeled $DD|_{I_1}$ in the lower diagram connecting points G and H . $DD|_{I_1}$ is read as “the DD curve given that $I = I_1$.”

Figure 20.3 DD Curve Effects from a Decrease in Investment Demand



Now, suppose I falls to I_2 . The reduction in I leads to a reduction in AD, ceteris paribus. At the exchange rate $E_{\$/\text{£}}^1$, the AD curve will shift down to $AD(..., E_{\$/\text{£}}^1, I_2, ...)$, intersecting the forty-five-degree line at point K . Point K above, which corresponds to the combination $(E_{\$/\text{£}}^1, I_2)$, is transferred to point K on the lower diagram. This point lies on a new DD curve because a second exogenous variable, namely I , has changed. If we maintain the investment level at I_2 and change the exchange rate up to $E_{\$/\text{£}}^2$, the equilibrium will shift to point L (shown only on the lower diagram), plotting out a whole new DD curve. This DD curve is labeled $D'D'|_{I_2}$, which means “the DD curve given is $I = I_2$.”